

Marcus Ruth

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EDUCATION

University of Central Florida
Bachelor of Science in Computer Science

Aug. 2024 – May 2027
Orlando, FL

Valencia College
Associates of Arts - General Studies

Aug. 2022 – May 2024
Orlando, FL

RELEVANT EXPERIENCE

University of Central Florida

Jan 2025 – Present

Teaching Assistant - Data Structures & Algorithms (COP 3502C) and Machine Learning (CAP 4611)

Orlando, FL

- Lead lab sessions and office hours covering data structures (trees, heaps, hash tables) and ML topics including classification pipelines, ensemble methods, and clustering.
- Designed programming assignments including a BST problem with augmented subtree sizes and a recursive permutation problem on team optimization.
- Grade assignments and provide constructive feedback to help students improve algorithmic problem-solving and debugging of C-based implementations.

Mindhome

December 2022 - February 2023

Software Engineer - Internship

Remote

- Completed ROS2 training utilizing Python, building foundational skills in AI and robotics programming.
- Contributed to backend development in Ruby, implementing API endpoints and assisting with database integration to support core platform functionality.
- Participated in code reviews and collaborated with senior engineers to refine application logic and improve system reliability.
- Developed a holistic understanding of software development methodologies including Agile workflows and version control practices.

PROJECTS

Cap Checker | *Next.js, TypeScript, Gemini API, Supabase, Playwright* | *BloomKnights 2026 Hackathon*

- Built a web app that fact-checks short-form financial videos, turning a pasted or uploaded clip into a source-backed scorecard that rates each claim as supported, contradicted, or unverifiable.
- Used Google's Gemini AI to pull spoken claims from videos and Google Search plus live Finnhub market data to verify them, then computed a deterministic "Cap Score" for each video.
- Built a searchable feed of vetted videos backed by Supabase, with streamed real-time analysis progress and full unit and end-to-end test coverage via Vitest and Playwright.

Terminal Presence | *Go, Discord IPC, systems programming*

- Built a cross-platform CLI tool that displays which command-line program you are actively using as a live Discord status, similar to how Discord shows the game you are playing.
- Detects 48 built-in tools by scanning running processes, spotlighting the busiest one with its logo, an elapsed timer, and a list of other active tools, all with no bot or token required.
- Supports hot-reloaded TOML configuration, privacy-first defaults, custom user-defined tools, and start-at-login on macOS and Linux; distributed via Homebrew and prebuilt Linux packages.

miniredis | *C++, POSIX sockets, systems programming*

- Built a Redis-like in-memory key-value store from scratch using modern C++ and POSIX sockets.
- Implemented event-driven TCP server with poll() capable of handling multiple concurrent client connections.
- Developed RESP protocol serialization and core commands (SET, GET, PING, ECHO) with per-client buffering.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Go, TypeScript, JavaScript, SQL

Web: Next.js, React, tRPC, Zod, Supabase, Postgres

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Nixtla

Developer Tools: Git, VS Code, Unix/Linux, AWS, PostgreSQL, Neovim